Visualize Lambda function invocations using AWS X-Ray

You can use AWS X-Ray to visualize the components of your application, identify performance bottlenecks, and troubleshoot requests that resulted in an error. Your Lambda functions send trace data to X-Ray, and X-Ray processes the data to generate a service map and searchable trace summaries.

Lambda supports two tracing modes for X-Ray: `Active` and `PassThrough`. With `Active` tracing, Lambda automatically creates trace segments for function invocations and sends them to X-Ray. `PassThrough` mode, on the other hand, simply propagates the tracing context to downstream services. If you've enabled `Active` tracing for your function, Lambda automatically sends traces to X-Ray for sampled requests. Typically, an upstream service, such as Amazon API Gateway or an application hosted on Amazon EC2 that is instrumented with the X-Ray SDK, decides whether incoming requests should be traced, then adds that sampling decision as a tracing header. Lambda uses that header to decide to send traces or not. Traces from upstream message producers, such as Amazon SQS, are automatically linked to traces from downstream Lambda functions, creating an end-to-end view of the entire application. For more information, see Tracing event-driven applications in the *AWS X-Ray Developer Guide*.

> **Note**
>
> X-Ray tracing is currently not supported for Lambda functions with Amazon Managed Streaming for Apache Kafka (Amazon MSK), self-managed Apache Kafka, Amazon MQ with ActiveMQ and RabbitMQ, or Amazon DocumentDB event source mappings.

To toggle active tracing on your Lambda function with the console, follow these steps:

To turn on active tracing

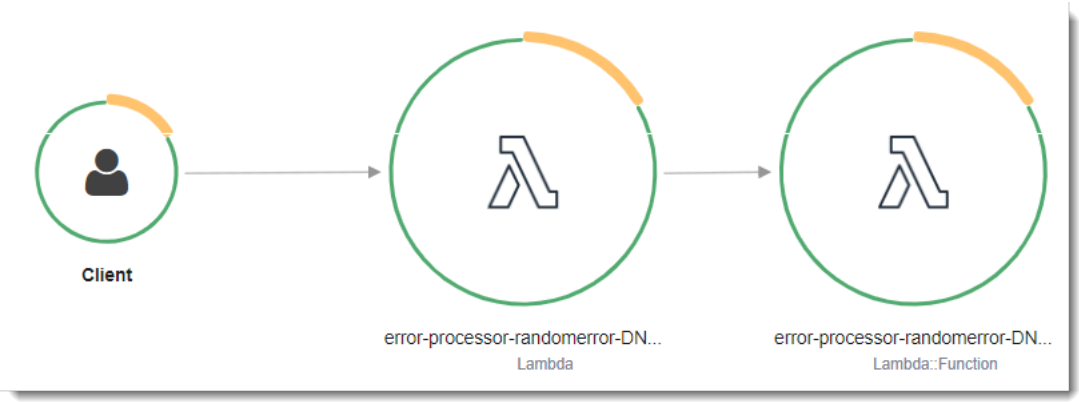
1. Open the Functions page of the Lambda console.
2. Choose a function.
3. Choose **Configuration** and then choose **Monitoring and operations tools**.
4. Under **Additional monitoring tools**, choose **Edit**.
5. Under **CloudWatch Application Signals and AWS X-Ray**, choose **Enable** for **Lambda service traces**.
6. Choose **Save**.

Your function needs permission to upload trace data to X-Ray. When you activate tracing in the Lambda console, Lambda adds the required permissions to your function's execution role. Otherwise, add the AWSXRayDaemonWriteAccess policy to the execution role.

X-Ray doesn't trace all requests to your application. X-Ray applies a sampling algorithm to ensure that tracing is efficient, while still providing a representative sample of all requests. The sampling rate is 1 request per second and 5 percent of additional requests. You can't configure the X-Ray sampling rate for your functions.

Understanding X-Ray traces

In X-Ray, a *trace* records information about a request that is processed by one or more *services*. Lambda records 2 segments per trace, which creates two nodes on the service graph. The following image highlights these two nodes:



The first node on the left represents the Lambda service, which receives the invocation request. The second node represents your specific Lambda function.

The segment recorded for the Lambda service, `AWS::Lambda`, covers all the steps required to prepare the Lambda execution environment. This includes scheduling the MicroVM, creating or unfreezing an execution environment with the resources you have configured, as well as downloading your function code and all layers.

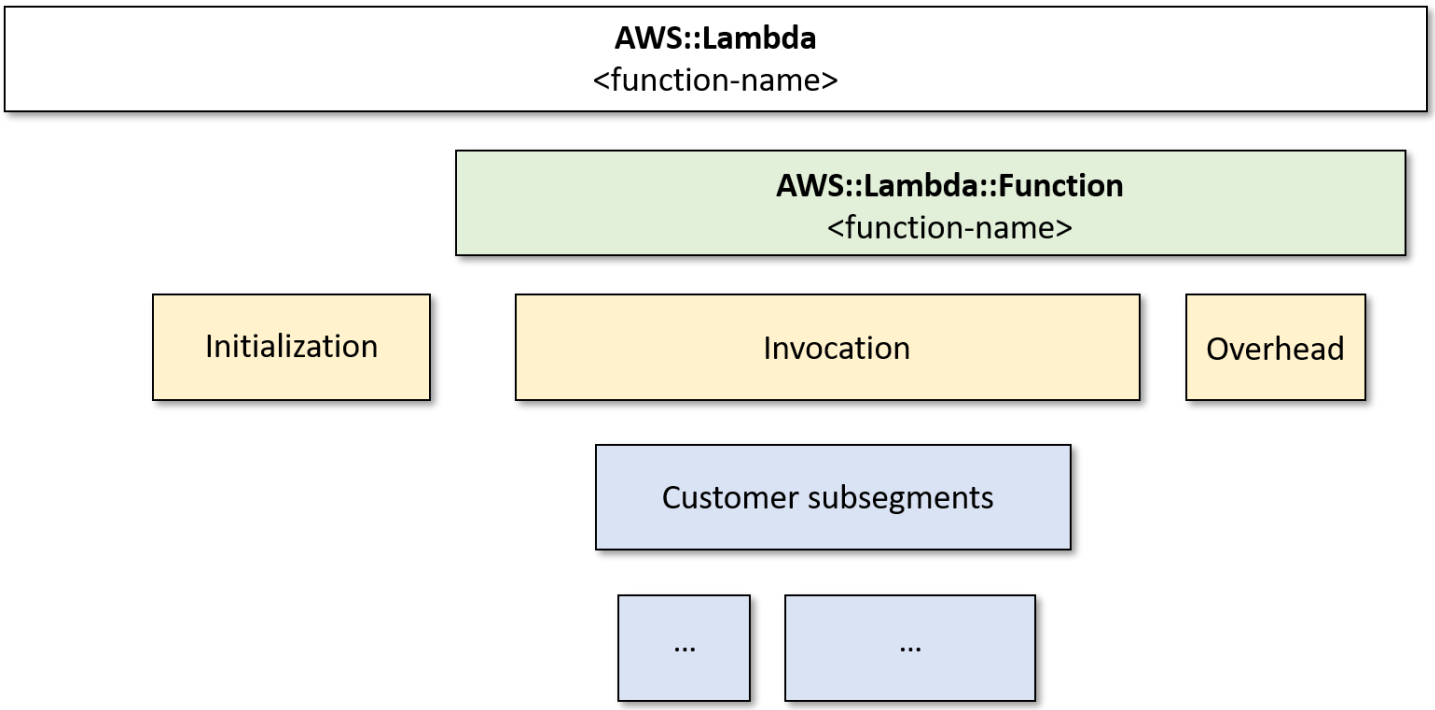
The `AWS::Lambda::Function` segment is for the work done by the function.

> **Note**
>
> AWS is currently implementing changes to the Lambda service. Due to these changes, you may see minor differences between the structure and content of system log messages and trace segments emitted by different Lambda functions in your AWS account.

This change affects the subsegments of the function segment. The following paragraphs describe both the old and new formats for these subsegments.
These changes will be implemented during the coming weeks, and all functions in all AWS Regions except the China and GovCloud regions will transition to use the new-format log messages and trace segments.

Old-style AWS X-Ray Lambda segment structure

The old-style X-Ray structure for the `AWS::Lambda` segment looks like the following:



In this format, the function segment has subsegments for `Initialization`, `Invocation`, and `Overhead`. For Lambda SnapStart only, there is also a `Restore` subsegment (not shown on this diagram).

The `Initialization` subsegment represents the init phase of the Lambda execution environment lifecycle. During this phase, Lambda initializes extensions, initializes the runtime, and runs the function's initialization code.

The `Invocation` subsegment represents the invoke phase where Lambda invokes the function handler. This begins with runtime and extension registration and it ends when the runtime is ready to send the response.

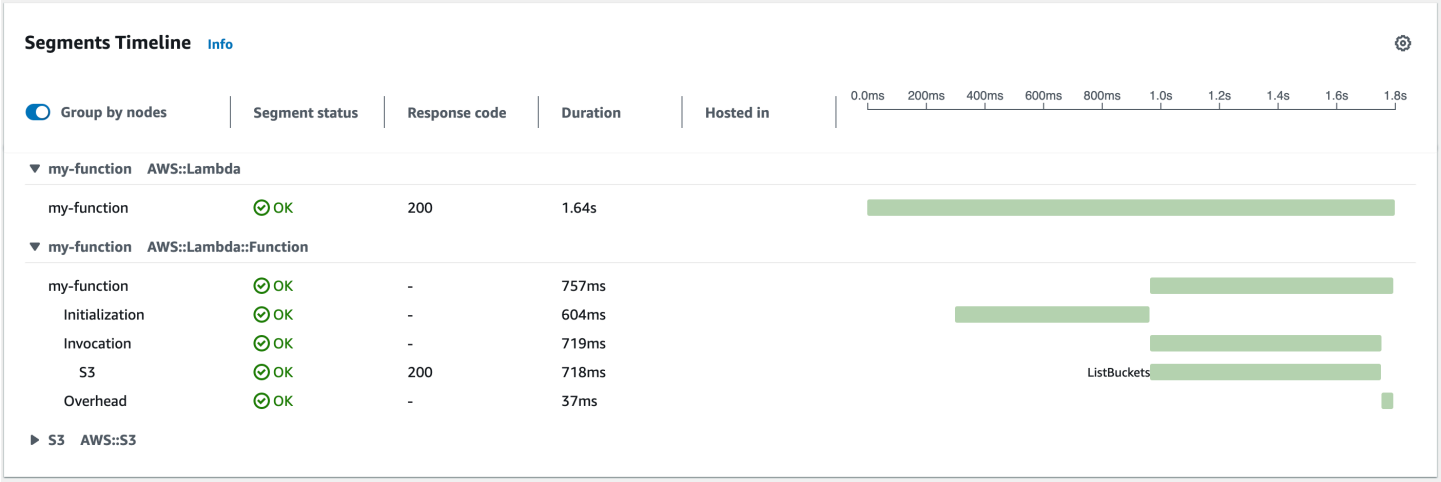
(Lambda SnapStart only) The `Restore` subsegment shows the time it takes for Lambda to restore a snapshot, load the runtime, and run any after-restore runtime hooks. The process of restoring snapshots can include time spent on activities outside the MicroVM. This time is reported in the `Restore` subsegment. You aren't charged for the time spent outside the microVM to restore a snapshot.

The `Overhead` subsegment represents the phase that occurs between the time when the runtime sends the response and the signal for the next invoke. During this time, the runtime finishes all tasks related to an invoke and prepares to freeze the sandbox.

> **Important**
>
> You can use the X-Ray SDK to extend the `Invocation` subsegment with additional subsegments for downstream calls, annotations, and metadata. You can't access the function segment directly or record work done outside of the handler invocation scope.

For more information about Lambda execution environment phases, see the section called "Execution environment".

An example trace using the old-style X-Ray structure is shown in the following diagram.

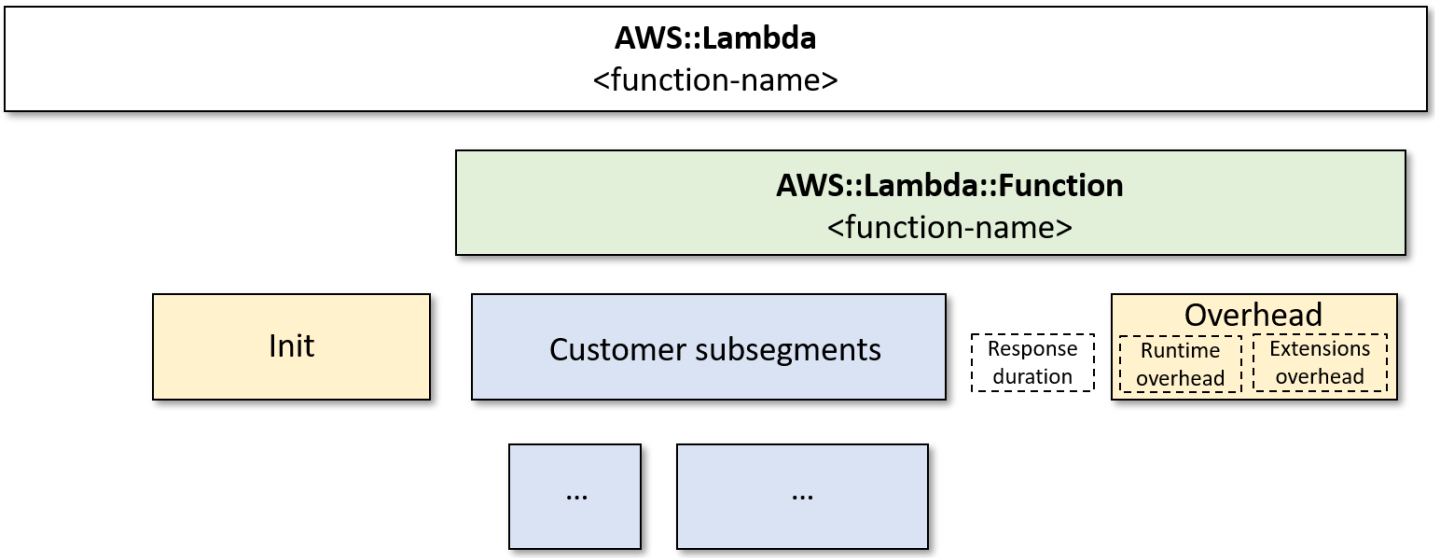


Note the two segments in the example. Both are named **my-function**, but one has an origin of `AWS::Lambda` and the other has an origin of `AWS::Lambda::Function`. If the `AWS::Lambda` segment shows an error, the Lambda service had an issue. If the `AWS::Lambda::Function` segment shows an error, your function had an issue.

> **Note**
>
> Occasionally, you may notice a large gap between the function initialization and invocation phases in your X-Ray traces. For functions using provisioned concurrency, this is because Lambda initializes your function instances well in advance of invocation. For functions using unreserved (on-demand) concurrency, Lambda may proactively initialize a function instance, even if there's no invocation. Visually, both of these cases show up as a time gap between the initialization and invocation phases.

New-style AWS X-Ray Lambda segment structure

The new-style X-Ray structure for the `AWS::Lambda` segment looks like the following:

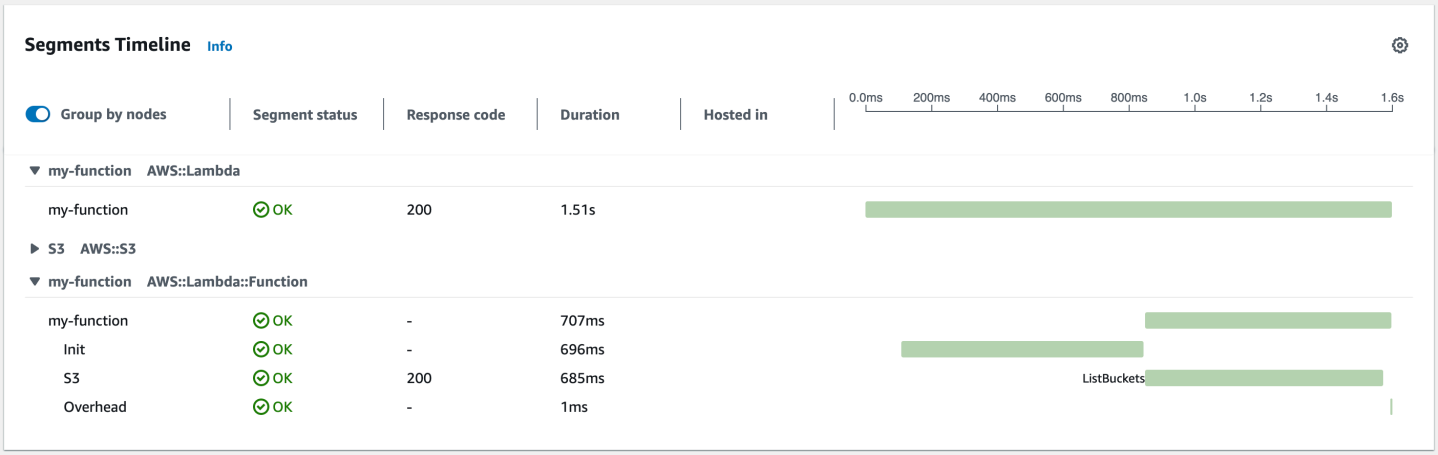


In this new format, The `Init` subsegment represents the init phase of the Lambda execution environment lifecycle as before.

There is no invocation segment in the new format. Instead, customer subsegments are attached directly to the `AWS::Lambda::Function` segment. This segment contains the following metrics as annotations:

- `aws.responseLatency` - the time taken for the function to run
- `aws.responseDuration` - the time taken to transfer the response to the customer
- `aws.runtimeOverhead` - the amount of additional time the runtime needed to finish
- `aws.extensionOverhead` - the amount of additional time the extensions needed to finish

An example trace using the new-style X-Ray structure is shown in the following diagram.



Note the two segments in the example. Both are named **my-function**, but one has an origin of `AWS::Lambda` and the other has an origin of `AWS::Lambda::Function`. If the `AWS::Lambda` segment shows an error, the Lambda service had an issue. If the `AWS::Lambda::Function` segment shows an error, your function had an issue.

See the following topics for a language-specific introduction to tracing in Lambda:

- Instrumenting Node.js code in AWS Lambda
- Instrumenting Python code in AWS Lambda
- Instrumenting Ruby code in AWS Lambda
- Instrumenting Java code in AWS Lambda
- Instrumenting Go code in AWS Lambda
- Instrumenting C# code in AWS Lambda

For a full list of services that support active instrumentation, see Supported AWS services in the AWS X-Ray Developer Guide.

Default tracing behavior in Lambda

If you do not have `Active` tracing turned on, Lambda defaults to `PassThrough` tracing mode.

In `PassThrough` mode, Lambda forwards the X-Ray tracing header to downstream services, but does not send traces automatically. This is true even if the tracing header contains a decision to sample the request. If the upstream service does not provide an X-Ray tracing header, Lambda

generates a header and makes the decision not to sample. However, you can send your own traces by calling tracing libraries from your function code.

> **Note**
>
> Previously, Lambda would send traces automatically when upstream services, such as Amazon API Gateway, added a tracing header. By not sending traces automatically, Lambda gives you the control to trace the functions that are important to you. If your solution depends on this passive tracing behavior, switch to `Active` tracing.

Execution role permissions

Lambda needs the following permissions to send trace data to X-Ray. Add them to your function's execution role.

- xray:PutTraceSegments
- xray:PutTelemetryRecords

These permissions are included in the AWSXRayDaemonWriteAccess managed policy.

Enabling `Active` tracing with the Lambda API

To manage tracing configuration with the AWS CLI or AWS SDK, use the following API operations:

- UpdateFunctionConfiguration
- GetFunctionConfiguration
- CreateFunction

The following example AWS CLI command enables active tracing on a function named **my-function**.

```
aws lambda update-function-configuration --function-name my-function \
--tracing-config Mode=Active
```

Tracing mode is part of the version-specific configuration when you publish a version of your function. You can't change the tracing mode on a published version.

Enabling `Active` tracing with CloudFormation

To activate tracing on an `AWS::Lambda::Function` resource in an CloudFormation template, use the `TracingConfig` property.

Example function-inline.yml – Tracing configuration

```
Resources:
  function:
    Type: AWS::Lambda::Function
    Properties:
      TracingConfig:
        Mode: Active
      ...
```

For an AWS Serverless Application Model (AWS SAM) `AWS::Serverless::Function` resource, use the `Tracing` property.

Example template.yml – Tracing configuration

```
Resources:
  function:
    Type: AWS::Serverless::Function
    Properties:
      Tracing: Active
      ...
```